

Joshua O'Callaghan

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EXPERIENCE

Senior Software Engineer

April 2025 - Present

Walmart

- Leading full-stack (Python + React) development of an AI-powered chatbot for inventory systems handling 5,000+ stores. Fine-tuned LLM models to optimize for user interaction and response precision.
- Mentoring junior engineers, promoting code review best practices, and driving architectural improvements.
- Reduced yearly database costs by \$400K+, optimized workloads across Cassandra, Cosmos DB, and Azure.

Software Engineer - Cloud

June 2023 - April 2025

Intel

- Optimized Kafka workload for client, doubling throughput while reducing CPU utilization by 20%.
- Improved compression workloads for clients, achieving a 12x increase in compression pipeline speed.
- Automated CI/CD pipeline and environment provisioning using Docker and Terraform; contributed to GitHub Actions workflows across AWS, Azure, and GCP.
- Ensured all systems were vulnerability-free while reducing cloud costs by 6% year over year.

Software Engineer - Media Processing

May 2021 - June 2023

Intel

Chandler, AZ

- Automated benchmarking and data collections to improve efficiency by 500%.
- Performed media analysis across various cloud/physical machines and optimized bottlenecks upwards of 80%.
- Designed backend and dashboards enabling real-time performance visibility for customers and internal teams.
- Led a team of engineers and stakeholders to deliver competitive media analysis.

Software Engineer

Sept 2019 - May 2021

Intel

Chandler, AZ

- Assembled various Python and Bash scripts for machine fab automation.
- Visualized data for error analysis with tools such as Plotly, and Tableau, reducing issue resolution time by 70%.

Software Engineering Intern

June 2019 - Aug 2019

American Express

Scottsdale, AZ

- Redesigned data infrastructure, enabling 2 teams with seamless data integration.
- Gained exposure to financial domain systems, administering Kubernetes cluster for data handling and uptime.

Software Engineering Teaching Assistant

Jan 2019 - June 2019

Arizona State University (CSE 340)

Tempe, AZ

SKILLS

Languages Proficient Python (6yrs) Intermediate Java (3yrs) React C/C++ Beginner Javascript Ruby on Rails

Software AWS, GCP, Azure, Terraform, Ansible, Docker, Kubernetes, Elasticsearch, Git, Bash

Soft Skills Leader, Communicative, Problem Solver, Innovator, Detail Oriented, Always Learning

PROJECTS

Disciple Assistant | Personal Project (~80 hours)

Dec 2024

- Used JS, YT pub sub, YT API, ChatGPT API and SQL to deliver discipleship automation for Impact Church.

Weather Dashboard | Personal Project (~50 hours)

Oct 2024

- Used Python, Elasticsearch, Logstash, Kibana and OpenWeather API to create an automated real-time dashboard.

Discord Bot Development | Personal Project (~30 hours)

Mar 2024

- Used Python, Java, React, Firebase DB, Discord API, ChatGPT API to create management bots for a discord server.

EDUCATION

Arizona State University, Ira Fulton School of Engineering

Master of Computer Science

Excelling Coursework: Data Structures and Algorithms, Advanced Algorithms, Distributed Databases

ACTIVITIES

ML Data Engineer - Evaluated coding performance on Claude, ChatGPT, Gemini, and other LLMs.

July 2024